



Asset Performance:

Rsl 01 Flood risk assessment



Credits



Minimum Standard

Aim

To encourage the identification of flood risk and implement mitigation measures where required.

Question

If a flood risk assessment (FRA) has been carried out, what was its scope and which flood risk level was assigned to the asset?

Credits	Answer	Select a single answer from B, C, E & F, if answer B or C has been selected, answer D is also available. Answer G is in addition to all other answers
0	A.	Question not answered
1	B.	FRA scope included Fluvial (rivers) and Tidal (sea) sources and flood risk from these sources is Medium or High
2	C.	FRA scope included all sources and flood risk is Medium or High
+2	D.	Flood mitigation measures were required and have been implemented
3	E.	FRA scope included Fluvial (rivers) and Tidal (sea) sources flood risk from these sources is zero or low
4	F.	FRA scope included all sources and flood risk is zero or low
Exemplary	G.	The Flood Risk Assessment contains an allowance for climate change

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	Flooding from the following sources must be taken into account: a) Fluvial (rivers) b) Tidal (sea) c) Surface water: sheet run-off from adjacent land (urban or rural)	C, F, G

Criterion	Assessment criteria	Applicable Answer
	d) Groundwater: most common in low-lying areas underlain by permeable rock (aquifers) e) Sewers: combined, foul or surface water sewers f) Reservoirs, canals and other artificial sources g) A nearby functional flood plain.	
2.	Flood risk maps produced by a Local or National Authority can form the basis of a Flood Risk Assessment but will not be sufficient on its own to demonstrate compliance. Assessors should verify which sources of flooding flood risk maps cover when awarding credits.	B, C, E - G
3.	Flood Risk Assessments need to be undertaken by a relevant organisation/authority or a competent individual: a) They should state which sources of flooding the report covers b) The report should have been written or reviewed in the last 5 years c) The level of detail required will depend on the size of the site and the arrangement of buildings on that site. For a small site (<2,000m ²) with a relatively simple arrangement of buildings this might consist of a brief report. For larger sites (>10,000m ²) with a higher density of buildings a more detailed assessment would be appropriate d) They should include flood mitigation measure if required.	B, C, E - G
4.	It must be demonstrated that recommendations have been implemented.	D
5.	Exemplary level credit: Answer option G can be select independently of which other answer option has been selected. An allowance for climate change should be based on a Medium or High Emissions Scenario from a robust Climate Model. This should cover a reasonable time period for the building, such as the 2050s or beyond.	G

Checklists and Tables

No National definitions of Flood Risk

In countries that do not have a definition of flood risk the following definitions can be used.

Table 25: Flood risk zones

	Fluvial (rivers)	Tidal (sea)
Low annual probability of flooding	Less than 1 in 1000 chance of flooding (< 0.1%)	Less than 1 in 1000 chance of flooding (< 0.1%)

	Fluvial (rivers)	Tidal (sea)
Medium annual probability of flooding	Between 1 in 100 and 1 in 1000 chance of flooding (1%–0.1%)	Between 1 in 200 and 1 in 1000 chance of sea flooding (0.5%–0.1%)
High annual probability of flooding	1 in 100 or greater chance of river flooding (>1%)	1 in 200 or greater chance of flooding from the sea (>0.5%)

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
1, 2	Flood risk maps showing the following: a) Location of the building b) Flood risk levels c) Sources of flooding covered
3	FRA from relevant body or competent individual, including evidence of their competence
4	Photographic or documentary evidence that the mitigation measures have been implemented
5	Highlighted sections of the FRA showing the Emissions Scenario, Climate Model used and timeframe for the allowance for climate change

Definitions

Competent individual:

An individual (or individuals) with qualifications and experience relevant to assessing flood risk, designing flood prevention measures and calculations. Where complex flooding calculations and prevention measures are required, this must be a specialist hydrological engineer.

Functional flood plain:

A zone that comprises land where water has to flow or be stored in times of flood.